Power Before Hours: Facility Limits, Embodied Carbon, and the Redundancy Dividend in Carbon-Aware Workload Shifting

Naveen Kumar Lakkaram · research summary · November 2026

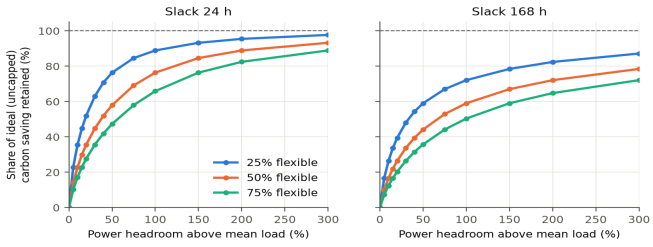
Question. Carbon-aware schedulers move flexible cloud work to cleaner hours and regions. Upper-bound studies (Sukprasert, Souza et al., EuroSys'24) optimise each job in isolation, implicitly letting all shifted work pile into the same clean hours. A real data centre's grid draw is capped by its electrical plant (utility service, transformers, switchgear, UPS, generators). How much of the carbon saving survives that cap, what is one extra kilowatt of capacity worth, and is it worth building?

Approach

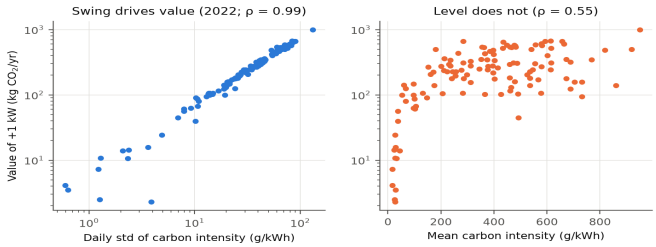
A clean-room reproduction of the EuroSys'24 analysis (123 grid zones, Electricity Maps 2022) matches **27 of 30** numeric claims, with three documented discrepancies. The core tool is an exact power-capped scheduling LP. Because all work shares one slack, earliest-deadline-first equals FIFO, so feasibility reduces to cumulative bounds: one variable per hour, proved equivalent to the per-arrival LP. Its dual variables are the *shadow carbon price of capacity*. Data: Google's public power traces (57 power domains), EIA-930 marginal signals, manufacturer EPDs, and LLM measurements on an Apple M4 Pro GPU.

Key findings

- **Power caps bound the savings.** The ideal temporal saving needs the median facility to draw 13x its mean power for an hour. +20% headroom keeps 36% of the ideal; at real Google headroom (median utilisation 70%, +42%) about 54% survives.
- **A unifying theory: capped flexible load is a lossless battery.** The LP is exactly a battery of energy slack × ϕ, discharge power ϕ and charge power h, so the optimum is water-filling the cleanest hours. Closed forms follow, with no optimiser: the share of the saving kept depends only on headroom per unit of flexible load, h/ϕ (exact); saving $\approx \phi \cdot A \cdot \mathrm{sinc}(\pi\phi/(\phi+h))$ for daily swing A (R^2 0.97 against the LP); and the value of a kW is a daily water-level area (R^2 0.98), which is linear in the swing and blind to the mean intensity. It also explains why UPS batteries barely help, and why only losses stop a scheduler serving both average and marginal ledgers: a lossless battery on a blended signal harms 0% of BAs, versus 60% at 90% round trip. Priced against a real 4 h battery, a half-flexible facility at +20% headroom is worth 4.3 kWh of storage per kW of load, and that battery would embody at least 16x the carbon of the headroom.
- **Build power, not servers.** One extra kW is worth 228 kg CO2/yr (median zone, +20% headroom). Grid-only electrical headroom repays its embodied carbon within its life in 98% of zones; CPU servers bought to absorb shifted work, in only 23%. The ordering held in every one of 5,000 Monte Carlo draws.
- **Redundancy is free carbon headroom.** Using the N+1 / 2N spare path for sheddable work closes 36% / 73% of the gap to the ideal, at zero embodied cost; maintenance and failures erode almost none of it.
- **Signal choice matters (extends e-Energy'24).** Across 48 US balancing authorities (EIA-930), the swing law holds for marginal signals (ρ = 0.96), but a capped schedule tuned to average intensity captures only 7% of the marginal-optimal saving and raises marginal emissions in 44% of them.
- **Under real caps, spatial and temporal flexibility compete.** In Europe, temporal shifting alone saves 5.4% but adds only 0.8 percentage points on top of migration, because migration fills the clean regions' headroom.
- **For LLM inference, batch first.** Measured on an M4 Pro GPU, batching cuts energy per token 4.4x; making batch size follow the grid adds at most 4.8% more.



Share of the ideal saving kept as power headroom grows.



Value of +1 kW tracks the grid's daily swing (left), not its level (right).

Why it matters and next steps

Carbon-aware schedulers should see watts, not only hours. The natural next step is an Ecovisor-style interface that exposes both the carbon signal and the facility's power budget to applications. Further steps: validate the capped model with real facility telemetry, repeat the GPU study on data-centre accelerators with power capping, and design carbon-optimal headroom that weighs the embodied carbon of the electrical plant against operational savings. My background in data-centre power equipment (UPS, switchgear, generators) is what led to this facility-side view.

Artifact: github.com/lakkaramnaveen/carbonshift. It includes tested open-source code (88 tests), all data, an interactive web explorer, a full paper and two workshop drafts. Every number above is generated from the result files.